

Exploring the efficient and hedonic shopping: A Comparative Study of in-game VR Stores

Yang Zhan^{*†}
Waseda University

Yiming Sun^{*‡}
Waseda University

Tatsuo Nakajima[§]
Waseda University

ABSTRACT

Shopping in Virtual Reality (VR) become popular in recent years since it provides immersive experiences. However, there is insufficient understanding of how efficient and hedonic features in VR stores affect the user experiences concurrently. This work aimed to address this gap by integrating a 2D user interface store and a 3D diegetic store into a VR game for comparative analysis. We explore the effects of efficient and hedonic factors on the users' perceptions and experiences. Results from a within-subject study ($N=14$) revealed that the diegetic store surpasses the 2D store in offering hedonic features, providing suggestions for VR store designs.

Index Terms: Human-centered computing—User studies; Virtual reality; User interface design

1 INTRODUCTION

Shopping is an essential part of daily life, where consumers not only purchase products but also fulfill their psychological needs, including personal and social motives [1]. While contemporary online stores offer the convenience of remote shopping, they often fall short in delivering vivid product presentations, immersion, and social experiences. The advent of VR shopping, to some extent, mitigated these limitations, prompting numerous studies to explore user experiences in VR stores, including shopping mechanisms and psychological effects [9]. Although these studies have enhanced the comprehension of VR shopping, they predominantly treated the VR stores as a novel form of online retail, focusing on physical goods. However, the VR stores dealing in virtual goods, like the in-game stores, also play a vital role in daily usage scenarios, which are overlooked by previous studies. The user experiences within the VR in-game stores remain inadequately understood. Furthermore, previous studies concentrated primarily on the target of finishing the purchase and neglecting the hedonic factors inherent in the shopping process. A high-quality shopping experience necessitates the holistic consideration of usability and satisfaction. Therefore, there is a lack of study on VR stores considering the efficient and hedonic factors at the same time within the in-game environment.

In our work, we explore the users' perception of efficient and hedonic factors, and how these two factors affect the user experiences. We started from two prevalent in-game store designs, namely the 2D user interface and 3D diegetic store, and then conducted a within-subject study ($N = 14$) to uncover the different user experiences. A VR shooting game (Fig. 1C) was designed to serve as the contextual background and shopping motivations. Our findings confirmed that the 3D store outperformed the 2D store in hedonic factors, offering better shopping and gaming experiences. Our work offers insights and suggestions for the future designs of VR in-game stores.

^{*}These authors contributed equally to this work.

[†]e-mail: yang.zhan@akane.waseda.jp

[‡]e-mail: ysun60@toki.waseda.jp

[§]e-mail: tatsuo@dcl.cs.waseda.ac.jp



Figure 1: Screenshots of our VR stores and game. A) shows the 2D store. B) shows the 3D store. C) shows the game scene.

2 RELATED WORK

Much research has been dedicated to improving user experiences and comprehending user behavior in VR stores. One type of study aims to optimize usability. Speicher et al. [5] studied the benefits of offline and online stores and proposed a speech-based VR store. An apartment metaphor was further designed to compare the forms of the shopping cart and the manipulation method [6]. However, these study cases set time constraints on purchase tasks, which cannot fully simulate the daily shopping process. While certain research delves into users' perceptions during shopping, encompassing workload [8], immersive experiences [3], and purchase impulses [2], the existing body of research concerning hedonic factors remains insufficient.

3 EXPERIMENTAL DESIGN

Our study, focused on user experiences in virtual object stores, necessitates relevant background for context and shopping motivation. Thus, a VR shooter game was designed, linking in-game store items to the game's thematic elements. The goal of the game is to eliminate all the enemies located within a designated room. Users can teleport around the room to either locate or evade the enemies and are equipped with guns for shooting. All pertinent operations are executed through controllers. Related sound effects are integrated to remind the enemy respawns and being attacked.

We incorporated two types of stores into our game. The first, a 2D store (Fig. 1A), features a large panel displaying item information and a purchase button. The second, a 3D store (Fig. 1B), emulates a realistic store environment, allowing users to move around and interact with various items. To delve into the social aspect of shopping, we introduced a multiplayer feature, prompting participants to communicate during the shopping process. In the 2D shop, players can engage in voice chat; in the 3D store, they can see each other's avatars¹ while undertaking shopping tasks. Reflecting the game's shooting genre, the stores offer items like weapons and potions, along with attribute-enhancing goods that, for instance, boost a player's attack value.

3.1 Measures and Procedure

We evaluated our work by conducting a within-subject study. Our evaluation is based on two pieces of questionnaires. One is the short version of the User Experience Questionnaire (UEQ-S), which contains limited items but still measures the product qualities in the

¹<https://developer.oculus.com/documentation/unity/meta-avatars-overview/> (accessed on 27 November 2023)

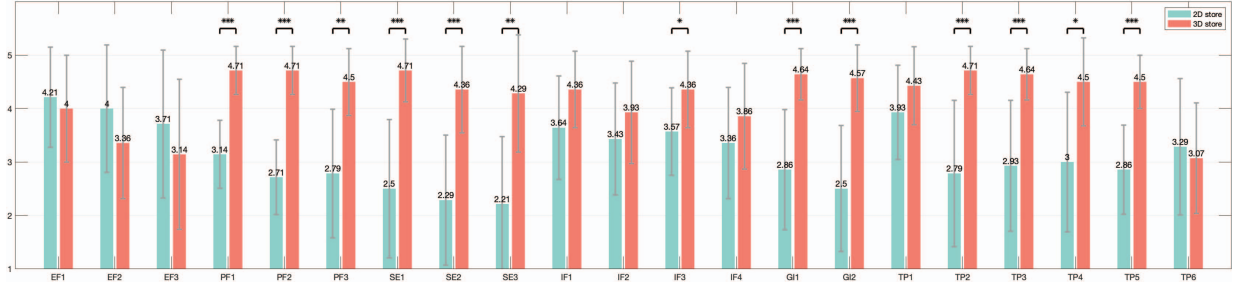


Figure 2: The results of our post-experimental questionnaire, encompassing means and standard deviations across six sub-scales. Statistical significance level are denoted by * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$).

same way as the full version of the UEQ [4]. The second questionnaire, developed by our team, comprises six sub-scales: Efficiency (EF), Playfulness (PF), Social Experience (SE), Informativeness (IF), Game Integration (GI), and Telepresence (TP). For each sub-scale, we formulated several questions to effectively measure their respective aspects. To guarantee the questionnaire's reliability, the measure of TP was adapted from the Presence Questionnaire [7].

For the user study, we recruited a total of 14 participants ($M = 24.3$, $SD = 2.63$) from the university to join our experiment, comprising 13 males and 1 female. Among them, 42.9% have no prior experience using a VR headset, 42.9% have used VR less than once a month, and 14.3% use a VR headset on a weekly basis. The following outlines the experiment procedure: For each batch of the experiment, we invited two people to participate. Two participants will try the same store demo together. Upon completing each demo, they will be asked to complete the two pieces of the questionnaire. The two stores were conducted in alternating sequences to mitigate the influence of the demo sequence on experimental outcomes.

4 RESULTS

In this section, we present the results of two sets of questionnaires. We used the Wilcoxon Signed-rank test to analyze statistical significance since all the data were ordinal.

	2D Store	3D Store	p-value
PQ	1.179	1.125	$p > 0.1$
HQ	-0.875	2.375	$p < 0.001$
Overall	0.152	1.750	$p < 0.001$

Table 1: The scaled mean results of the UEQ-S sub-scales and significant analysis for the 2D and 3D store.

Table 1 displays the results from our UEQ-S questionnaire, which assesses the pragmatic quality (PQ) and hedonic quality (HQ) of our product. The scoring for both attributes spans from -3 to 3, with 'overall' indicating the mean score of these attributes. Both 3D and 2D stores scored positively in PQ, each exceeding a score of 1, suggesting above-average PQ for both. In contrast, while the 3D store excelled in HQ with a remarkable score of 2.375, the 2D store lagged with a negative score of -0.875. Figure 2 displays the results from our custom questionnaire. As the figure shows, the 3D store received significantly higher scores compared to the 2D store in the playfulness category. This trend is also observed in the SE, GI, and TP categories, where the 3D store consistently receives higher ratings across all SE and GI sub-questions. Furthermore, in the TP category, four out of six sub-questions show significant score differences between the 2D and 3D stores. Under the IF class, although a significant difference was observed only in the IF3 sub-category, the 3D store also received higher overall ratings in the

general trend of the IF category.

5 CONCLUSION

In our study comparing 2D user interfaces with 3D diegetic stores in games, the UEQ-S results reveal that while the 2D store is marginally more efficient in terms of PQ, the 3D store significantly excels in HQ. Our custom questionnaire corroborates these findings, showing comparable efficiency in both store formats but highlighting the 3D store's superior joyfulness and immersion. Moreover, it seamlessly integrates into the game's environment, and offers a better social experience, aligning with the UEQ-S results to demonstrate its effectiveness in enriching the gaming experience.

REFERENCES

- [1] M. J. Arnold and K. E. Reynolds. Hedonic shopping motivations. *Journal of retailing*, 79(2):77–95, 2003. doi: 10.1016/S0022-4359(03)00007-1
- [2] J. V. Chen, Q.-A. Ha, and M. T. Vu. The influences of virtual reality shopping characteristics on consumers' impulse buying behavior. *International Journal of Human-Computer Interaction*, 39(17):3473–3491, 2023. doi: 10.1080/10447318.2022.2098566
- [3] A. Schnack, M. J. Wright, and J. L. Holdershaw. Immersive virtual reality technology in a three-dimensional virtual simulated store: Investigating telepresence and usability. *Food Research International*, 117:40–49, 2019. doi: 10.1016/j.foodres.2018.01.028
- [4] M. Schrepp, A. Hinderks, and J. Thomaschewski. Design and evaluation of a short version of the user experience questionnaire (ueq-s). *International Journal of Interactive Multimedia and Artificial Intelligence*, 4:103, 2017. doi: 10.9781/ijimai.2017.09.001
- [5] M. Speicher, S. Cucerca, and A. Krüger. Vrrshop: A mobile interactive virtual reality shopping environment combining the benefits of on-and offline shopping. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 1(3):1–31, 2017. doi: 10.1145/3130967
- [6] M. Speicher, P. Hell, F. Daiber, A. Simeone, and A. Krüger. A virtual reality shopping experience using the apartment metaphor. In *Proceedings of the 2018 International Conference on Advanced Visual Interfaces*, pp. 1–9. Association for Computing Machinery, 2018. doi: 10.1145/3206505.3206518
- [7] B. G. Witmer and M. J. Singer. Measuring presence in virtual environments: A presence questionnaire. *Presence: Teleoperators and Virtual Environments*, 7(3):225–240, 1998. doi: 10.1162/105474698565686
- [8] N. Xi, J. Chen, F. Gama, M. Riar, and J. Hamari. The challenges of entering the metaverse: An experiment on the effect of extended reality on workload. *Information Systems Frontiers*, 25(2):659–680, 2023. doi: 10.1007/s10796-022-10244-x
- [9] N. Xi and J. Hamari. Shopping in virtual reality: A literature review and future agenda. *Journal of Business Research*, 134:37–58, 2021. doi: 10.1016/j.jbusres.2021.04.075